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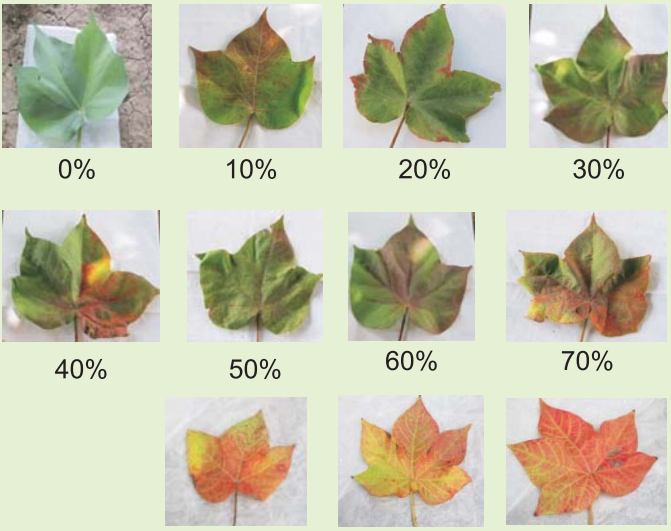
Technical Bulletin 30

# Leaf Reddening and its Management in cotton

## Leaf reddening Intensity




**National Centre for Integrated Pest Management**
LBS Building, Pusa Campus, New Delhi
&
Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola

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# Leaf Reddening and its Management in cotton

R. M. Gade[1], R.K. Tanwar[2] , P. Jeyakumar[2] and V. Kanwar[2]

भाकृअनुप
ICAR

**National Centre for Integrated Pest Management**
LBS Building, Pusa Campus, New Delhi[2]
&
Dr. Panjabrao Deshmukh Krishi Vidyapeeth, Akola[1]

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***Authors***

R. M. Gade[1], R.K. Tanwar[2] , P. Jeyakumar[2] and V. Kanwar[2]
[1]DR PDKV Akola
[2] NCIPM, New Delhi

***Cover page photographs***

Cotton field affected with leaf reddening
Leaf reddening intensity

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Dr. C. Chattopadhyay
Director
National Centre for Integrated Pest Management
LBS Building, Pusa Campus, N. Delhi- 110012
Tel.:91-011-25843935, 25843936, 25740951, 25740952, 25843985
Fax: 91-011-25841472, E-mail: ipmnet@bol.net.in Web:www.ncipm.org.in

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## Introduction

Red leaf disease of cotton, commonly known as *lal patti* in India (Fig. 1), Bronze wilt, "copper top," "sudden wilt" and "phloem wilt" in the U.S. and "red leaf," "red wilt" and "anthocyanosis" in other countries, is a physiological disorder induced by different abiotic stresses. Bronze wilt as a problem in cotton was identified in 1995 in Mississippi and Louisiana, and 1996 in Texas, Arkansas, Tennessee and North Carolina and 1997 in Georgia and California. Recently, this has emerged as a serious problem in various cotton growing belts of Pakistan especially in Sindh. In India, though the leaf reddening was identified long back yet the problem has acquired the status of major disease only after the cultivation of Bt cotton hybrids.

Figure 1: Red leaf disease of cotton

This physiological disorder is the outcome of the interaction of variety/ hybrid, environmental factors and availability of nitrogen including the micronutrients. Generally it is observed in poor soils with low nitrogen (N) supply, fertilizer dose with inadequate N supply, water stress due to poor drainage in soil hampering the N viability and N absorption, and sudden reduction in night temperature arresting the translocation and high radiation.

Devastating losses due to Bronze wilt was reported in the upper Mississippi River Delta and in the Gulf Coast States in 1998. In India various degrees of losses in seed cotton yield has recently been observed. Lint quality is also adversely affected because of the premature crocking of the bolls which ultimately results in low price of their produce in the market.

## Symptoms

The common symptoms of leaf reddening include:

- Leaf margins turn yellow and later red pigmentation is formed on the whole leaf area
- Reddening progresses to stems and leaves
- Shedding of leaf and boll and the poor plants growth
- Fewer bolls and in extreme cases no bolls and wilting of the plant

Figure 2: Common symptom of leaf reddening-upper part of the leaf raised giving rough appearance

Symptoms generally appear during boll development and may become severe as bolls mature. Symptoms usually progress through a series of stages. First prominent symptom is the appearance of red colour on the first fully expanded leaf near the top of the plant. The area between the veins in the upper part of the leaf becomes raised, giving the leaf a rough appearance (Fig. 2). The stems turn red to the top of the plant, appertaining similar to that following herbicide application. Absence of flowers or squares is the another characteristic symptom but the affected plants can bear bolls. In case of sever incidence of leaf reddening, all the foliage may rapidly wilt and collapse within a few days of the first symptoms followed by darkening and narcotizing the leaf and stem tissue of the upper canopy. Peduncles of bolls along with branches, leaves subtending bolls, and the main stem are most often affected.

## Incidence and intensity

For scoring the disease incidence five spots (four in the corner and one in the centre) are selected in a field of one acre and at each spot 20 plants (total plants and plants with red leaf symptoms) are counted. Based upon the observation, recorded at five spots, the per cent leaf red plants are calculated. For scoring intensity, a system of 0-9 grade has been developed which is based upon the leaf area covered with red colouration (Table 1; Fig.3).

**Table 1: Leaf reddening grading depending on leaf area covered with reddening**

| Grade | Leaf covered with reddening |
|---|---|
| 0 | No leaf reddening |
| 1 | 1 or less than 1% leaf area covered |
| 3 | 1 to 10% leaf area covered |
| 5 | 11-25 % leaf area covered |
| 7 | 26-50 % leaf area covered |
| 9 | More than 50 % leaf area covered |

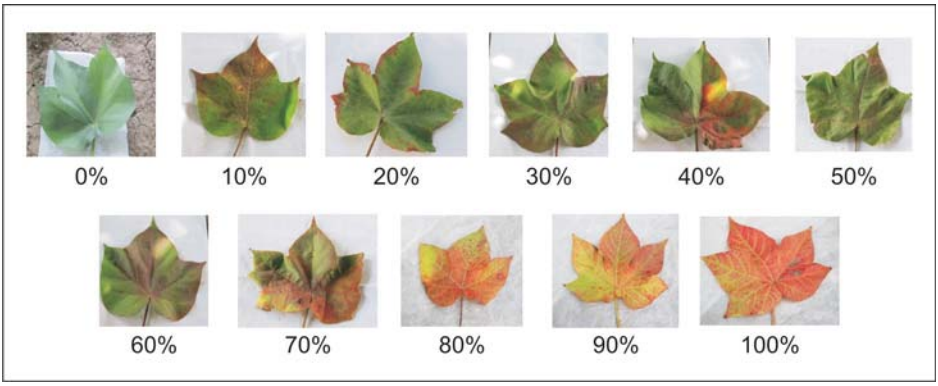


Fig. 3: Leaf reddening Intensity

## Trend of leaf Reddening

Data collected on the incidence of leaf reddening under National Information System for Pest Management (NISPM) project of TMC MMII at 14 centres located in 9 states indicated the presence of leaf reddening in most of the project districts with variation in the incidence. The trend of leaf reddening (2010-11) indicated that disease incidence progressively increased from 35 SMW onwards at Ahmednagar and Jalna (Maharashtra), Khandwa (Madhya Pradesh), Anand and Rajkot (Gujarat) and Belgaum (Karnataka) and crossed 50% after 43 SMW, however, at Rajkot the disease incidence reached more than 80% in 38 SMW. During the season the highest incidence was recorded at Ahmadnagar (49 SMW) followed by Rajkot (38 SMW), Belgaum (47 SMW), Perambalur (1 SMW) Khadwa (50 SMW) and Anand (50 SMW). The lowest incidence was recorded at Guntur (Fig . 4). The incidence, in general, remained at low level in north zone as compared to central and south zones.

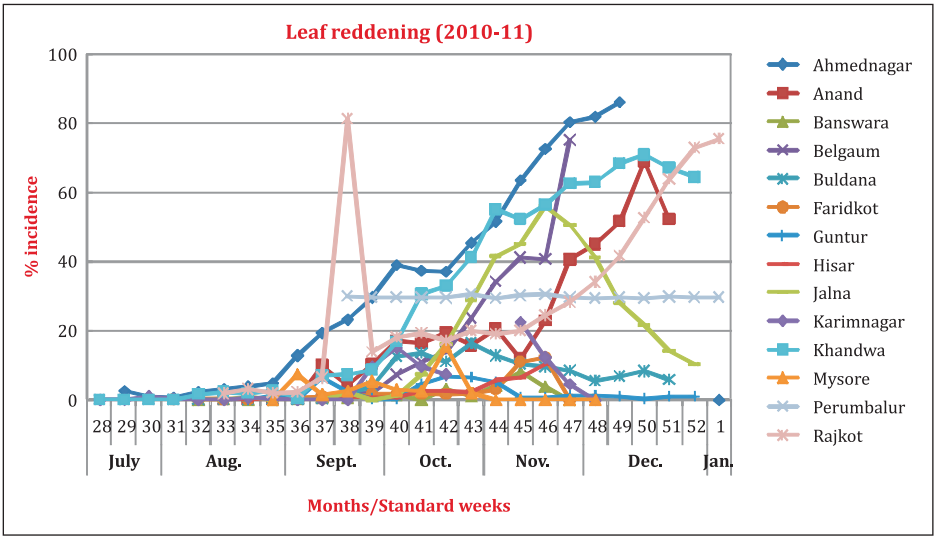


Fig.4: Trend of leaf reddening in cotton in 14 districts in 9 states

Studies conducted under the NISPM Project on the trend of leaf reddening during the last three years indicated the appearance of leaf reddening at all the 14 centres during 2010-11 and at 1968 occasions (Weekly observations recorded from 280 villages at 14 centres in 9 states) it also crossed the ETL. During 2011-12 and 2012-13 the incidence has shown declining trend as it was observed in 10 and 5 centres, respectively. It also crossed the ETL at 349 and 193 occasions during 2011-12 and 2012-13, respectively (Fig. 5).

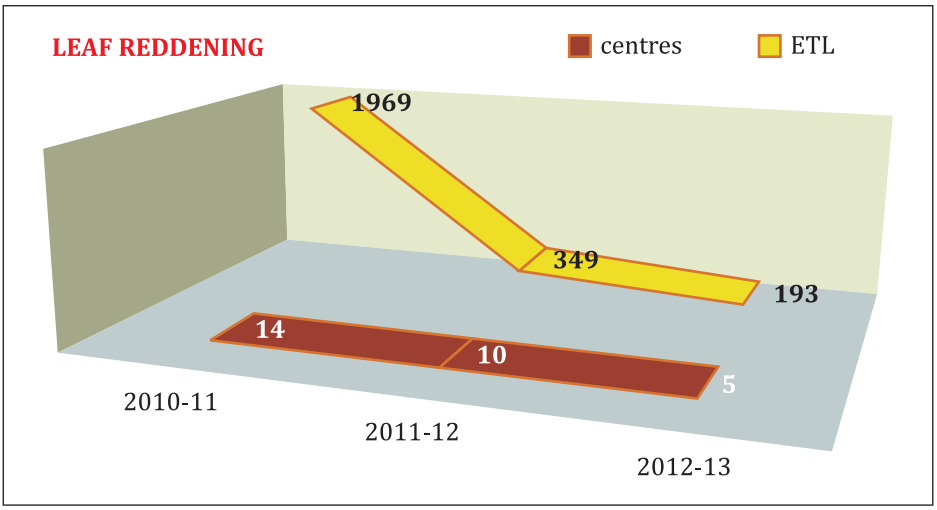


Fig.5: Appearance of leaf reddening at different centres (Nos.) and on the occasions (Nos.) it crossed the ETL

## Leaf reddening process

In the leaf reddening process colour of leaves generally changes from green to yellow and then to red, but in many cases the change is directly from green to red. The process of appearance of reddish shades in the normal green colour begins in an early phase of cotton growing; maxima expression of red colouration of both bottom and upper leaves is observed in adult plants at the end of vegetative period. During the leaf reddening the leaves lose turgidity, droop and wither away. Sometime the colour spreads from the epidermis to the adjacent parenchymatous cells. High tannin content and low cell sap of pH are prerequisite for development of red pigment in cotton leaf. Pigment has been suggested as an oxidative condensation product of tannic acid.

### Seasonal changes

During rainy days, particularly in August, because of the low light intensity and temperature, the cotton plant is exposed to low blue and red radiation that restricts the growth of plant. Nutrient uptake in the month of August is also affected due to heavy rainfall. During the primary stage, the condition of high temperature and high radiation prevailing in September impairs the cellular metabolism. Moreover, in September/ October the rainfall also ceases and days become warm. During the boll formation stage leaf nitrogen content gets drastically reduced due to translocation to the developing bolls. These conditions have been noted to lead to the onset of red leaf syndrome on cotton plants. In November, higher wind flow causes a gradual depletion of soil moisture levels. There is also wide difference in day and night temperature. High radiation enhances the photosynthesis rate, which brings deposition of carbohydrates in leaves. These multiple factors appear to create an ideal platform for the enhancing the leaf reddening process.

### Physiological and biochemical changes

The abnormal red colour of leaves might be due to accumulation of red pigments, called anthocynanis, which is accompanied by degradation of chlorophyll. Anthocyanin pigments are usually confined to the epidermal cell. The upper epidermis generally has more pigments than the lower. In red cotton leaves photosynthesis apparatus get reduced due to decreased photochemical activity and stomatal limitations. Increased synthesis of anthocyanins in vegetative plant parts has a protective role against various stresses stimuli. Anthocyanins interfere with oxidative stress induced by improper temperature, irradiance, and osmotic and high salinity constraints (Bohnert *et al.*, 1999; Verma, 1999).

Anthocynine content increased in reddening affected leaves in the range of 4.94 to 5.06% at square formation stage and at rest of the growth stages. Other plant components such as proloine, peroxidas and lipid peroxides also play important role in leaf reddening process (Fig. 6). Proline is a ubiquitous compatible solute and act as an osmoregulator. Prolonie has also been suggested to be involved in built up of cell wall proteins, balancing the ratio of NAD(P)+ /NAD (P)H (redox signalling), C and N storage as well as active oxygen scavenging (Hare *et al.*, 1998; Verma, 1999; Hasegawa *et al.*, 2000) .

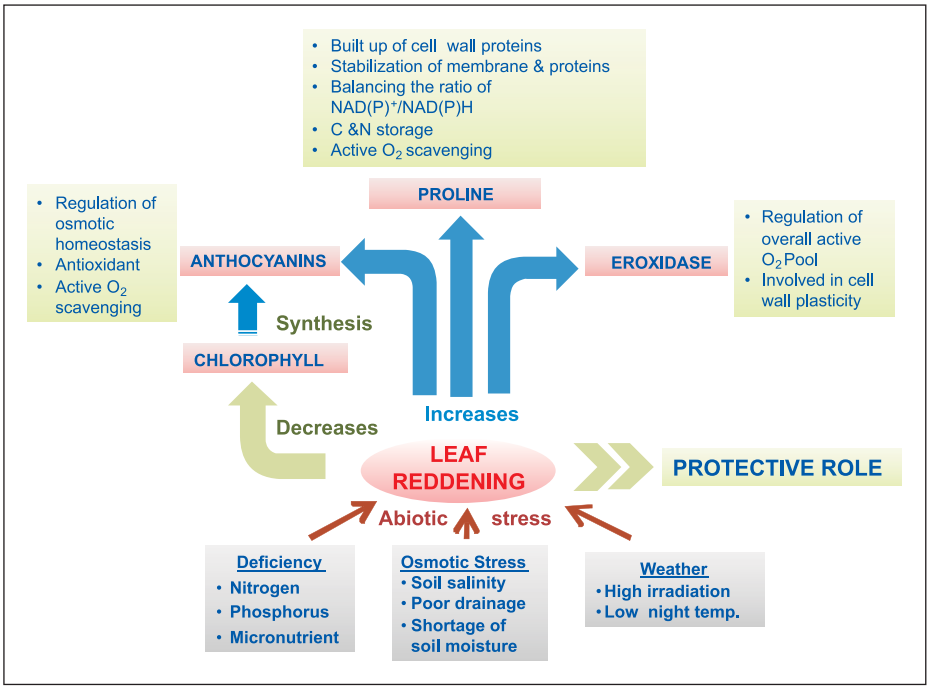


Fig. 6: Diagrammatic representation of leaf reddening process

Experiments conducted on cotton leaf reddening by Edreva *et al.*, (2002) registered low accumulation of anthocyanins in the stage of light symptoms and a dramatic (662%) increase in anthocyanins component in at severe symptoms of reddening. In contrast total Chlorophyll (a+b) content continuously declined. The content of proline revealed a tendency to decrease under mild stress (slight reddening) but at severe reddening an important accumulation of proline was observed. In reddening leaves the lipid peroxide content followed a trend to decrease from the stage of light symptoms to the stage of severe symptoms. The activity of peroxidase (PO) was strongly increased at the stage of light symptoms which further enhanced at severe reddening (Table 2). The

enhancement of PO activity in reddening leaves may also have a protective function. PO acting as a $H_2O_2$ scavenger is involved in the regulation of the overall active oxygen pool and cell signalling *via* $H_2O_2$. PO is important in plant responses to abiotic stresses including high salinity. The lipid peroxide content indicative of membrane fragmentation was decreased. This multicomponent system encompassing anthocyanins, proline, and peroxidase may act co-ordinately to overcome abiotic stress in cotton.

**Table 2: Contents of total antocyanins and chlorophylls, proline, lipid peroxides (mmol kg-1 (d.m.) and peroxidase (IEU) in cotton leaves with light and sever symptoms of reddening**

| Variants | Total antho-cyanins | | Total Chl (a+b) | | Proline (%) | | Lipid perox-ides | | Peroxidase | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | (%) | | (%) | | (%) | | (%) | | (%) |
| Controls | 1.00a | 100 | 3.20a | 100 | 10.70a | 100 | 0.28a | 100 | 20.04a | 100 |
| Light symp-toms of leaf reddening | 2.58b* | 258 | 2.03b* | 63 | 7.93b | 74 | 0.19b | 68 | 45.57b* | 227 |
| Severe symptoms of leaf red-dening | 6.62c* | 662 | 1.16c* | 36 | 27.28* | 255 | 0.16c | 57 | 77.53c* | 387 |

*Edreva et al., (2002)*

*Values in the same vertical column followed by different letters are significantly different at P<1% (*) and 5%*

### Histochemical changes

Electron microscopic study of cotton leaves carried out by Stoyanova-Koleva *et al*., 2005 established the osmiophilisation of the membrane compartment during leaf reddening. They observed massive agglomerations referred as electron dense osmiophilic substance (most probably of anthocyanin nature), on the tonoplast in the vacuole of cells under the adaxial epidermis. In chloroplast stroma a zone of low electron density enclosing numerous osmiophilic aggregations of unclear chemical character was also differentiated.

## Factors responsible for leaf reddening

Though the exact cause of the disorder has yet to be investigated but both biotic as well as abiotic factors have been proposed as possible causative agents. It is believed that high concentrations of certain strains of *Agrobacterium* spp.

(*Agrobacterium tumefaciens*), a soil and seed bacterium causing damage to the secondary roots, are responsible for of bronze wilt (http://texaserc.tamu.edu). Expression of the bronze wilt is also influenced by cultivar, environment (temperature, light intensity, and photoperiod), production practices and abiotic stresses (high temperature and water stress), soil type and fertility, relative nitrogen content of fertilizer and phosphorous/ potassium ratio in fertilizer. Potassium deficiency frequently has been mentioned as a primary suspect. Plant age, fruit set and load, earliness in fruiting and maturity, and high yield potential also effect symptom severity.

## Biotic factors

Among the sucking pests the jassids have been considered as most important insect acting as vector in carrying the red leaf disease. Reddening in cotton may also be due to a virus transmitted by *Aphis gossypii* or a mycoplasm leading to reduction in boll weight and yield. In certain cases attack of the *Bemisia tabaci* during the ball maturation period of the cotton crop has also resulted leaf reddening.

## Abiotic factors

- **Weather:** Temperature plays a major role in the regulation of anthocyanin production. High radiation results in faster photosynthesis which leads to deposition of carbohydrates in leaves. Accumulation of products of photosynthesis favours formation of chromogeans and red pigments. Mechanical injury and lowering the temperature cause the development of red pigment. Though the anthocyanin production is independent of light yet its production is favoured by light. Certain factors such as drought followed by rain or irrigation, sudden change in atmosphere causing excessive evaporation, cloudy weather and low night temperature (15-20$^{o}$C) during the flowering and boll development period (reducing the translocation), low temperature around 10 to 15$^{0}$ C and higher solar radiation, high wind velocity (causing gradual depletion in soil moisture levels due to decrease in the soil temperature and evapo-transpiration).

- **Soil:** Poor soil with less capacity to supply nitrogen and phosphorus leads to leaf reddening. Cotton is primarily grown on alluvial, black and red soils. The properties of this soil such as texture, depth, structure, crusting, fertility, water holding capacity and moisture characteristics have tremendous influence on cotton plants. In alluvial soil, crusting, rising water table, salinity and alkalinity are major problems whereas the black soil suffers from moisture scarcity, cracking and water permeability problem. Soil

condition such as high alkalinity, water stress because of poor drainage in soil (hampering N availability and N absorption), severe shortage in soil moisture (reducing the translocation), and high soil temperature are the factors which individually or in combination are responsible for leaf reddening. Close spacing of plants causes the deficiency of nitrogen resulting in stunted growth and reduced leaf size. It is proposed that 1.5% N in leaves may be taken as critical concentration for nitrogen deficiency in cotton (Bhatt *et al.*, 1984).

- **Mineral and nutrition deficiency:** In India, since cotton is growing on soils with 6.0 to 8.5 pH, therefore, the availability of Ca and Mg is not limiting. However, sometimes excess exchangeable Ca may lead to Mg deficiency. When the Mg content in the leaves at flowering stage, falls below 0.3 per cent, reddening of leaves is observed. Mg deficiency can be corrected by soil or foliar application of $MgSo_4$. Studies conducted on reddened leaf indicated that all major elements except Mg were found deficient in red leaf (Dhopte, 1990). Micronutrient contents of red leaf were also reported to be far below the sufficiency level. However, the reports are also available indicating occurrence of gradual reddening due to Mg deficiency in MCV-5 and CBS 158 cotton (Perumal and Subramaniam, 1979).
- **Hybrid/Variety Susceptibility:** Some of the *Gossypium hirsutum* varieties and a few inter and intra specific tetraploid hybrids are sensitive and vulnerable to this malady. Indigenous *G. arboreum* desi cottons, in general, thrive well under all stresses and are tolerant to red leaf syndrome. Currently, as the problem is emerging in most of the cotton growing states, therefore, there seems to be no Bt hybrid fully resistant to this disease.

## Distinguishing leaf reddening from other insect pests/ diseases damages

Leaf reddening is often confused with the symptoms of jassid attack or Magnesium deficiency

- Due to jassid attack, the leaves lost turgidity and turn yellow and later slightly reddish with a typical cup shaped turning of margins. Outbreak of jassids and aphids at the primary phase turns the young leaves red due to heavy feeding metabolism. Delay

Fig. 7 : Cotton leaf damaged by jassids

between two spray against these pests develop blue to deep pink coloration of the leaves which ultimately turns yellow leaves to red (Fig. 7).

- In case of Magnesium deficiency only the older leaves are affected and the symptoms appear from leaf margin. Moreover the Magnesium deficiency induces purplish red colour. In case of leaf reddening disorder, a normal young and old leaves suddenly turn to red without any symptoms of margin distortion. The attack instead of margin initiates from midrib (Fig. 8).

Fig. 8: Magnesium deficiency symptoms in cotton plant

- In case of *Fusarium* or *Verticillium* wilt yellow or brown streaks develop in and around xylem vessels in the wood of plants whereas the pith, wood and bark of the lower stem and upper tap root remain light colored and appear healthy with bronze wilt. Fungal root rots cause discoloration and rotting, first of the bark and then of the entire root (http://texaserc.tamu.edu.).

## Losses

Cotton losses in Georgia valued at nearly $25 million In 1998. Sporadic outbreaks have occurred each year in Missouri and Georgia, resulted in substantial reductions in yield, fiber quality and seed quality for some producers. In India due to red leaf disorder the cotton yields reduced from 10-50%. The percentage leaf reddening is negatively correlated with seed cotton yield. In certain cases no yield was obtained.

## Management strategies

Planting hybrids/varieties that have no known history of leaf reddening is the easiest way to reduce losses from this disease. In the absence of disease resistant varieties/hybrids, the immediate solution lies in the adoption of modern agricultural techniques and management practices. There is a need to take remedial measures if red leaf damage is more than 10-15% of the total crop. The following practices are useful in minimising the leaf reddening.

- Avoid planting the susceptible hybrids/varieties.
- Avoid repeated cultivation of cotton on same field as it may require manurial applications to build up sufficient organic carbon. Follow the crop rotation.

- Adjustment of sowing time for enabling the crop to skip over the adverse environmental (high soil temperatures) condition during boll development stage.
- Adequate drainage to avoid water logging of the fields.
- Provide mulching if reddening is due to drought conditions.
- Prevent water stress by irrigating sufficiently.
- Balanced and timely use of fertilizer.
- Foliar application of micronutrients, especially zinc, boron and copper at recommended rate and at specific growth stages.
- When soil tests indicate deficiencies, fertilize with phosphorus, potassium and sulfur before planting.
- Take remedial measures if damage is more than 10-15% of the total crop.
- Spray 2% DAPS or 2% urea at an interval of 8-10 days till the symptoms subsides.
- Soil application of DAP @ 50 Kg/ha helps in reducing the leaf reddening under irrigated conditions
- Application of 0.5% Mg $S0_4$ as spray had resulted 18-23% increase in yield and 50-60% recovery in greenness in MCV-5 cotton.

## Field trial on management of leaf reddening

Application of macro- and micro-nutrients in the form of aqueous sprays in external form could increase metabolic processes like photosynthesis, respiration, chlorophyll a/b ratio and reduced leaf reddening. Recently field trials were conducted by Gade *et al.*, (2013) during 2010-12 at Dr. PDKV, Akola on demonstration of management strategies for leaf reddening in Bt cotton at different locations in Maharashtra (Akola, Bhaurad, and Kanheri ) with five treatments (Table 3).

**Table 3: Details of different treatments**

| Treatment | Intervention |
|---|---|
| $T_1$ | Three sprays of 1 % $MgSo_4$ at pre flowering , flowering and boll setting stage |
| $T_2$ | Three sprays of 0.05 % $KNO_3$ at pre flowering , flowering and boll setting stage |
| $T_3$ | • Basal dose Humic Acid 50 Kg /ha, calcium nitrate 25 Kg/ ha, sulphur 25 Kg/ha<br>• After sowing Humic acid liquid 2.5 lit/ha/375 lit water + Calcium nitrate 2.5 Kg /ha/375 lit water + Sulphur 2.5 Kg /375 lit water<br>• Three applications of Humic acid liquid 2.5 lit/ha/375 lit water + calcium nitrate 2.5 Kg /ha/375 lit water + Sulphur 2.5 Kg /375 lit water at 21 days interval |
| $T_4$ | Three sprays of 2 % DAP at pre flowering , flowering and boll setting stage |
| $T_5$ | Control |

The results of the experiments are summarized below:

- **Red leaf intensity:** Results of the trial (pooled mean of three years) conducted on red leaf intensity at three locations indicated that all the treatments having micronutrients interventions ($T_1$, $T_2$, $T_3$ and $T_4$) were found significantly superior to control ($T_5$) (Fig.9). Among these four treatments, $T_1$ showed lowest per cent intensity of red leaf at 60 days at all the locations whereas at 90 and 120 days, the treatment $T_3$ also indicated lowest red leaf intensity at Bhourad and Kanheri (Table 4). Overall pooling of all the days and locations data exhibited lowest leaf reddening intensity in $T_1$ followed by $T_3$ (Fig. 10)

**Table 4: Red leaf intensity (%) recorded at three locations during 2009-10, 10-11 and 11-12 (Pooled mean)**

| Treatment | Bhourad | | | Akola | | | Kanheri | | |
|---|---|---|---|---|---|---|---|---|---|
| | 60 days | 90 days | 120 days | 60 days | 90 days | 120 days | 60 days | 90 days | 120 days |
| $T_1$ | 32.16 | 46.35 | 44.07 | 30.33 | 42.37 | 48.22 | 33.93 | 56.91 | 56.59 |
| $T_2$ | 37.47 | 45.69 | 48.03 | 37.59 | 49.55 | 52.71 | 37.37 | 62.36 | 64.88 |
| $T_3$ | 34.61 | 42.1 | 43.31 | 35.89 | 56.59 | 55.75 | 38.32 | 53.86 | 58.13 |
| $T_4$ | 35.47 | 44.84 | 50.01 | 35.53 | 54.06 | 52.89 | 36.59 | 64.3 | 63.01 |
| $T_5$ | 48.46 | 54.7 | 59.79 | 42.08 | 64.53 | 62.27 | 42.51 | 65.55 | 70.33 |

| Treat-ment | Bhourad | | | Akola | | | Kanheri | | |
|---|---|---|---|---|---|---|---|---|---|
| | 60 days | 90 days | 120 days | 60 days | 90 days | 120 days | 60 days | 90 days | 120 days |
| SE ± | 0.029 | 0.046 | 0.045 | 0.058 | 0.034 | 0.061 | 0.029 | 0.062 | 0.022 |
| CD at 5 % | 0.08 | 0.14 | 0.13 | 0.16 | 0.10 | 0.18 | 0.08 | 0.18 | 0.06 |

Fig.9: Expression of leaf reddening in control-$T_5$ (Left) and $T_1$ (Right)

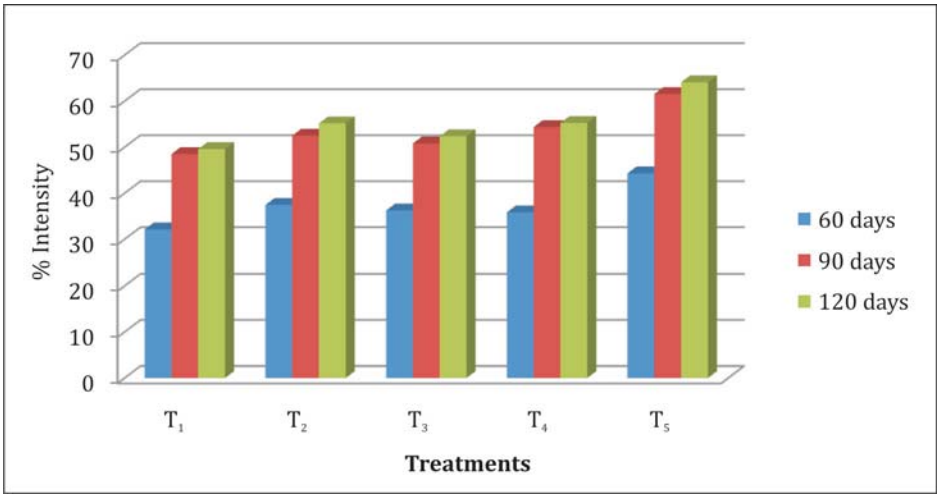


Fig. 10: Leaf reddening intensity in different treatments (Pooled data)

- **Chlorophyll:** Chlorophyll a and b remained significantly higher in all the treatments ($T_1$, $T_2$, $T_3$ and $T_4$) supplemented with nutrients as compared to control ($T_5$). Among the four treatments the highest total chlorophyll was recorded in treatment $T_3$ followed by T1 (Table 5).

**Table 5: Effect of different treatment on Chlorophyll contents during 2011-12**

| Treat-ments | Bhaurad | | | Akola | | |
|---|---|---|---|---|---|---|
| | Chloro-phyll a | Chloro-phyll b | Total Chloro-phyll | Chloro-phyll a | Chloro-phyll b | Total Chloro-phyll |
| $T_1$ | 0.79 | 0.41 | 1.33 | 0.69 | 0.31 | 1.07 |
| $T_2$ | 0.73 | 0.26 | 1.09 | 0.61 | 0.16 | 0.89 |
| $T_3$ | 0.83 | 0.46 | 1.37 | 0.74 | 0.36 | 1.25 |
| $T_4$ | 0.54 | 0.24 | 0.88 | 0.44 | 0.18 | 0.69 |
| $T_5$ | 0.38 | 0.15 | 0.67 | 0.32 | 0.12 | 0.54 |
| SE± | 0.06 | 0.11 | 0.14 | 0.05 | 0.02 | 0.11 |
| CD (P=0.05) | 0.17 | 0.32 | 0.41 | 0.14 | 0.05 | 0.31 |

- **Yield:** Highest yield was recorded in $T_3$ followed by $T_1$. In general, all the treatments provided with nutrients ($T_1$, $T_2$, $T_3$ and $T_4$) were superior to control $T_5$ (Table 6).

**Table 6: Effect of different treatments on yield (q/ha) during 2009-10, 10-11 and 11-12**

| Treat-ments | Bhourad | | | Akola | | Kaneri | | Pooled mean |
|---|---|---|---|---|---|---|---|---|
| | 2009-10 | 2010-11 | 2011-12 | 2010-11 | 2011-12 | 2010-11 | 2011-12 | |
| $T_1$ | 26.15 | 17.30 | 32.45 | 8.11 | 5.71 | 13.00 | 32.6 | 19.19 |
| $T_2$ | 25.70 | 16.92 | 29.19 | 7.46 | 5.37 | 12.16 | 30.67 | 18.21 |
| $T_3$ | 27.60 | 18.79 | 33.33 | 8.50 | 6.08 | 13.77 | 33.76 | 20.12 |
| $T_4$ | 23.40 | 17.00 | 29.31 | 7.72 | 5.17 | 12.39 | 28.25 | 17.61 |
| $T_5$ | 20.10 | 14.07 | 27.81 | 7.36 | 4.06 | 11.12 | 28.05 | 16.22 |
| SE ± | 3.60 | 0.42 | 1.48 | 0.06 | 0.67 | 0.32 | 1.29 | 0.37 |
| CD at 5% | 10.13 | 1.30 | 4.18 | 0.15 | 1.89 | 0.99 | 3.63 | 1.10 |

- **Economic analysis:** While calculating the overall expenditure, it was observed that in spite of higher yield in treatment $T_3$ the net monetary return was highest in $T_1$ followed by $T_2$ and $T_4$. The treatment $T_3$ has shown negative monetary return (Table 7).

**Table 7: Economic analysis of different treatments during 2009-10, 10-11 and 11-12 (Pooled mean) (In Rs.)**

| Treat-ments | q/ha | Gross Monetary return (Rs.) | Expenditure on red leaf man-agement (Rs.) | Net mone-tary return (Rs.) | Increase over con-trol (Rs.) |
|---|---|---|---|---|---|
| $T_1$ | 19.19 | 63327 | 1305 | 62022 | 8496 |
| $T_2$ | 18.21 | 60093 | 1098 | 58995 | 5469 |
| $T_3$ | 20.12 | 66396 | 20970 | 45426 | (-)8100 |
| $T_4$ | 17.61 | 58113 | 1099.08 | 57014 | 3488 |
| $T_5$ | 16.22 | 53526 | 0 | 53526 | 0 |
| SE± | 0.37 | | | | |
| CD(P=0.05) | 1.10 | | | | |

*Average rate of Seed cotton Rs.3300/q*

Based on the overall results of the experiment three sprays of 1 % $MgSo_4$ at pre flowering, flowering and boll setting stage were recommended to mange leaf reddening in cotton

## Important references


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